



Arts and Technology Week 2022

February 21-24, 2022

Programming Contest

Before you begin

All teams will be given the same problem that requires a software solution, develop it within the Arts and Tech Week 2022 duration, and present it at **February 22, 2022 3PM**

Teams can use any programming language i.e. SDK, or IDE, or other solutions to achieve their products.

Presentations are limited to 15 minutes per team, with 5 minutes Q&A from the panelist.

The Problem

An event organizer is planning to organize a large gathering for people that are interested in technology. To make this event viral, they want to have an invite system that allows one (1) invited person to invite two (2) more, just like how the [Katipunan Triangle System](#) works.

However, they do not want to go into the trouble of verifying each of the attendee's inviter during the event because it will be tedious and time consuming. To mitigate this, they thought of a code system (like QR or Barcode) that can be traced back from its inviter.

Your task now is to create one. Let me give you an example on how it should work.

Let's start with the topmost inviter - the organizer. Let's assign a Hexadecimal value **6E053DFE** as the inviter's code.

Now, the organizer invites Joanna and Alyanna. We can make their codes start with the last four (4) places of the organizer's code, **3DFE**. Joanna's code is **3DFE5BCB**, and Alyanna's will be **3DFEDBLE**.

Now, we can now determine Joanna's reference value, which is **5BCB**. Alyanna's reference value is **DBLE**.

The system should be able to generate two (2) invite codes if an inviter inputs his/hers, and the codes should be saved on the node. The same system should also verify if the invite code is valid or not.

A separate system should be created to present this linking system.

You can use any type of code values for each invitee (You can even use QR or barcodes).

A graphical user interface is not necessary but will make things a little bit fancier for the user.

Criteria for Judging

Relevance of the solution to the given problem	50%
Use of technologies in the solution	30%
Presentation Flow	20%
TOTAL	100%

How should the presentation be done?

It's up to you! Feel free to use any presentation style or medium to explain your solution.

The whole code doesn't need to be explained, except if there are questions from the panelist.

The presentation will be done in a Zoom Virtual meeting room on February 22, 2022, starting at 3PM.

Other stuff that I want to say

I won't be entertaining any questions after you have received this document.

Be resourceful.

Again, any programming language, tools, SDK, IDE, or technology can be used, as long as you put the effort in developing the solution.

GOOD LUCK!